

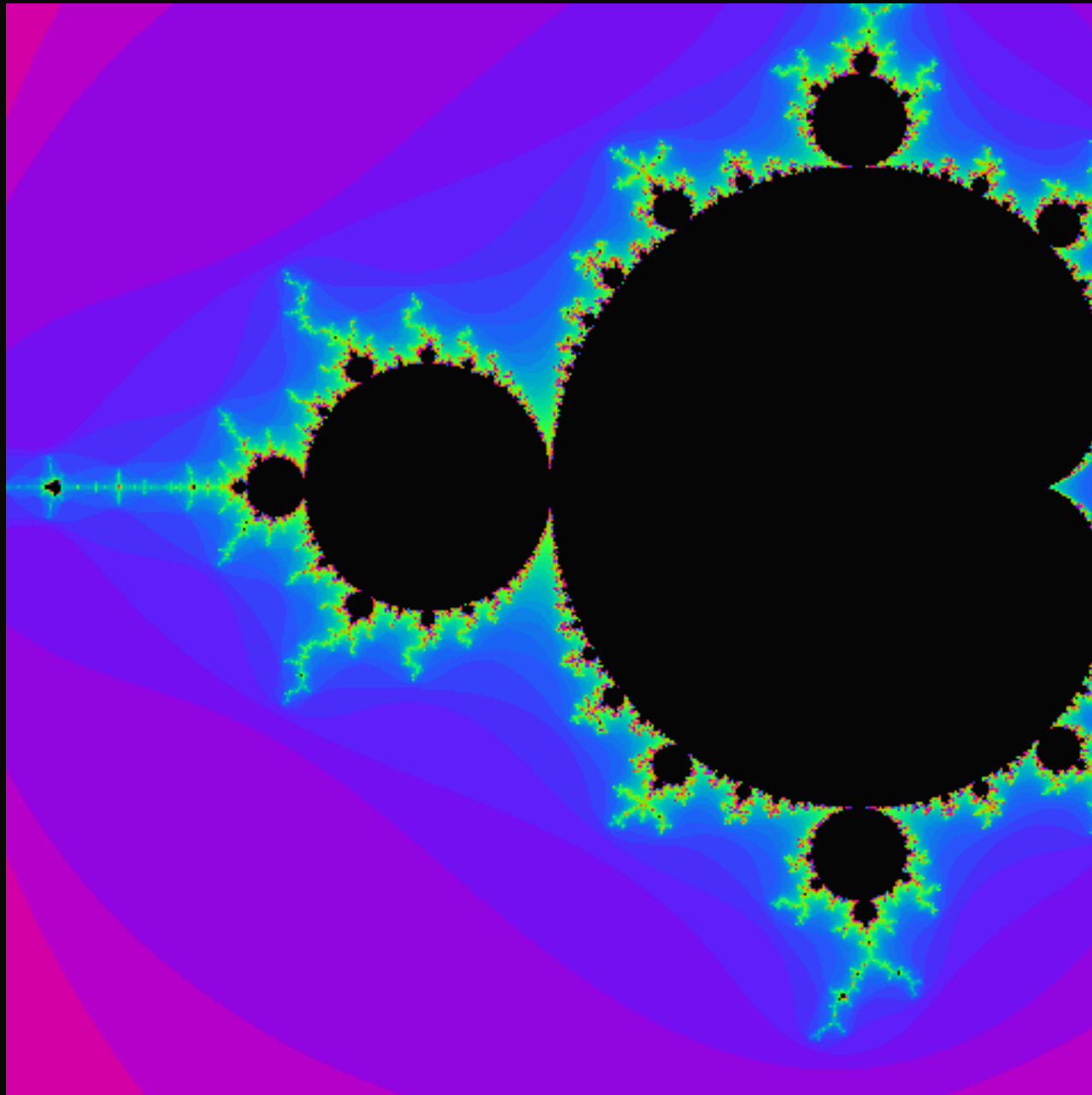
NO GPU · NO GRAPHICS LIBRARY · NO OS

We drew this on a computer with no operating system.

One program, 16 MiB of RAM, and a socket. Nothing else is running — there is nothing else to run.

The PNG is encoded by hand, because there was no image library to call.

FRAME 1, STRAIGHT OFF THE MACHINE



480x480 · palette PNG written byte by byte in C
this copy drawn by the Linux build of the same source – see slide 12

WHAT IS ACTUALLY UNDERNEATH

A unikernel. The program is the machine.

No kernel. No shell. No init, no container, no package manager, no language runtime. The hypervisor loads a 2.8 MB image and execution begins in `main()`.

So there is no libpng to link, no zlib, no framebuffer device. If you want a picture, you compute every pixel and you write the file format yourself.

Deflate has a mode that does nothing.

Porting zlib to call one function is the larger job. But a deflate **stored block** is a five-byte header wrapped around at most 65,535 literal bytes — no compression, no Huffman tree, no tables.

A zlib stream is a two-byte header, a run of those, and an Adler-32. Every decoder on earth accepts it.

CRC32 table, 20 lines. Encoder, about 80.
Validated against a real decoder, not just a viewer that opened it.

THE PART THAT IS NOT DECORATION

Every frame is rendered twice.

An escape count is a pure function of the pixel. Render the same frame twice and a correct machine returns the same image, byte for byte, *by construction*.

We had already measured this platform disagreeing with itself on fixed arithmetic — about **one result in four thousand**, with a Linux control under the same hypervisor coming back clean.

So: render twice, compare, paint every disagreeing pixel red. The picture is the measurement.

AND THE ANSWER WAS

0

**red pixels. Every frame,
every build.**

Which is a result, not a disappointment. The corruption we measured is in **64-bit integer** modular arithmetic — gcc lowers it to a **__udivti3** call.

A Mandelbrot loop is double-precision floating point, in SSE registers. Different path. So far, an uncorrupted one.

THEN IT WOULD NOT SEND

787 KB left the machine and never arrived.

```
[!] send: Failure when receiving data from the peer  
[!] frame 1 not delivered
```

The same binary, the same frame, posted from Linux on the same host: fine. So it was the unikernel's network stack, not the API at the other end.

WRONG CONCLUSION #1

"It's too big. I need a real compressor."

Confidently reasoned from a single failure. A fixed-Huffman deflate encoder, five to ten times smaller, about a day of work.

I did not write it. I measured the wall instead – which is the only good decision in this entire sequence.

WRONG CONCLUSION #2

A ladder said 256 KB. To the wrong server.

```
PROBE 262144 bytes ok HTTP 401  
[+] largest body: 256 KB
```

Measured against **our own service**, then generalised to a machine-wide ceiling. The next frame was 231 KB, comfortably under it, and failed anyway.

WRONG CONCLUSION #3

A second ladder said 16 KB.

```
PROBE 12288 bytes ok HTTP 400  
PROBE 16384 bytes FAILED (31s)
```

Pointed at Telegram this time. Better. Then a 5,008-byte frame failed too — and no size hypothesis survives that.

Size was never the variable. It had just been big enough, every time, to look like one.

9

bytes into every PNG, there is a zero.

The probe bodies were ASCII. The frames are binary.

`CURLOPT_POSTFIELDS` in this port stops at the first NUL — while Content-Length still promises the rest, so the server waits for data that never comes, then closes.

Every frame failed at every size. That is what "no size hypothesis survives" was telling me.

SIX LINES, AND IT FLEW

```
struct upload up = { g_body, g_body_len };  
curl_easy_setopt(h, CURLOPT_READFUNCTION, read_cb);  
curl_easy_setopt(h, CURLOPT_READDATA, &up);
```

Feed the body a chunk at a time instead of handing over a pointer and a length. It is also what curl's own documentation recommends for anything that is not a C string.

WRONG CONCLUSION #4, AND #5

Not a ceiling. Then: also a ceiling.

4,093 bytes failed where 10,160 had posted, so I called the ceiling imaginary and the failures intermittent.

With retries added, both turned out true. Below about 16 KB it works but **often needs a second attempt**. At 17,065 bytes and above it has failed every attempt, every time.

One fault was hiding the other, in both directions.

SO IT SIZES ITSELF

The program finds its own limit, by posting.

It renders at 64, posts it, steps up, and keeps stepping until one fails. The largest that survives becomes the size it runs at.

Not a number I hardcoded from a measurement that might go stale — a number the machine establishes about itself, every time it boots.

Today that is 96x96. If the port is fixed, it will find that out on its own.

WHAT THIS COST, AND WHAT IT BOUGHT

Five wrong answers. Two real faults.

A compressor is needed. 256 KB. 16 KB. No ceiling at all. Each was drawn from real data, and each stopped one measurement short of the thing that would have contradicted it.

What is actually there: **POSTFIELDS truncates at the first NUL**, deterministically — and a transport that is **flaky below 16 KB and refuses above it**.

Including the retraction that itself needed retracting. That is what the record looked like.

THE BUG THAT ONLY APPEARS IF YOU KEEP GOING

Frame 1 fitted. Frame 2 did not.

```
frame 1  320x320  14,240 bytes  delivered
frame 2  320x320  17,065 bytes  not delivered
```

Deeper zooms carry more detail, so they compress worse. A size chosen once from the simplest frame in a sequence is a size that stops working three frames later.

So it re-checks every frame, and steps down a rung when the picture outgrows what the link will carry.

AND THE COMPRESSOR, FINALLY

96x96	9,312	->	2,907	3.2x
256x256	65,792	->	10,308	6.4x
480x480	230,880	->	26,687	8.7x

Fixed-Huffman deflate with LZ77, written by hand. Bigger frames compress better: more flat interior, longer runs, and a repeated scanline becomes one match at distance width+1.

Dismissed early for a bad reason, then built for a good one. Every size validates against a real decoder.

THE RESPONSE TO AN UNRELIABLE LINK

Retry, and count how often you had to.

Not a smaller payload, and not a threshold to stay under. Four attempts with widening backoff, and the retry count printed in every frame's caption.

If the link is flaky, the honest thing is to say so in public, on each picture, rather than to quietly hide it behind a size that seemed to work.

IT IS RUNNING NOW

Zooming 1.35x a frame, into the seahorse valley.

github.com/tabibazar/baremetal-agent

MIT. The renderer, the hand-written PNG encoder, the probe, the deploy scripts.

Built on Return Infinity's BareMetal. All three bugs reported, each with a reproducer.